INPUT

- App allows me to write the notes there. Use of BlockNote tool for Notion look and feel.
    - Recording the **timestamps** of when i typed, relative to the actual madrid time
    - Every time the teacher goes to the next slide, i place "//" divider to indicate it. Save it as a timestamped slide-transition event; it can be mapped to the uploaded presentation later.

- Upload of the rest of the sources once i'm done with class:

- manual selection of the subject through drop down selection.
        - ability to "+ new subject" button and name a new subject
        - widget appears to upload guia docent. using Schwartz-Hearst Algorithm, anatomic filtering, suffix and low freq word filtering, create a small, static glossary of words stored into that subject. Ensure that there are no duplicates.

- upload can either be: notes + presentation, notes + presentation + audio, notes + audio.
    - pdf support +60 pages (presentation) with extraction of text. ignore images.
    - support audio files (around two audio files) with total combined length of around 2 hours.
    - if audio is attached, enter recording start time. For a continuous recording, clock time = recording start + elapsed audio time.
        - if Start time unknown: optionally align one recognizable recording moment with a corresponding slide marker. Example: audio 04:20 matches the 09:14:20 marker, so recording start is approximately 09:10.

font end: TBD

ATOMIZATION

- transcription
    - ffmpeg to cut each AUDIO file in natural pauses every around 2 minutes = CHUNKS. About two minutes is an initial setting to test, not a fixed requirement; use speech detection if classroom noise defeats silence detection.
    - code cuts the chunk into the SENTENCES, chronologically, enumeration considers total sentences.
        = A01 = audio recording 1
        = A01-C005 = segment 5 of audio 1
        = A01-T0042 = sentence 42. this is the **unique ID** for transcriptions.

Sentence ID	Parent chunk
A01-T0042	A01-C006
A01-T0043	A01-C006
A01-T0044	A01-C006

- notes
    - each note block will have a **unique ID** based off simple enumeration
    - it's writing first edit / writing time stamp is recorded, as well as the last edit. (last edit will always be during class)

- presentation
    - each slide gets enumerated (S013). the slide will also get a respective time stamp which is given matching the presentation to the notes ("//").
        - the notes in // between // will have edges: the beginning note block and the final one. the timestamp for the slide uses the beginning noteblock's timestamp and the final one.
    - each bullet point from each slide gets extracted and enumerated (S013-B02) this will be the **unique ID** for each bullet point info.

AUDIO TRANSCIPTION

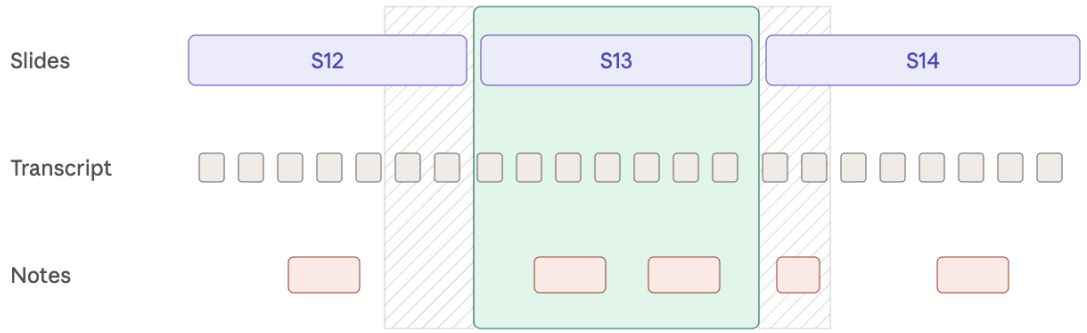
(*if audio is provided)*

- each transcription chunk has the respective timestamp (will be a range). The timestamp is clock-time, using the audio file creation time + the time of recording.
- the notes that match the timestamp range of the chunk are retrieved to create a mini glossary, filtering for:
        - anatomic words, low freq spanish words, medical suffixes
        - code caps the glossary at the selected provider's prompt-token limit. Groq currently documents 224 tokens.
- whisper large v3 transcribes one chunk at a time using personalized prompt to increase accuracy. estimated cost per 2 hours = 0.22 cents
    - request quality medata

- audio clean up:
    - regex noise filtering, "ya lo teneis?" "ya esta copiado?"... deterministic
    - anything that has been flagged - Provide one-click playback of flagged text with a few seconds of surrounding audio, plus manual correction.

POOLING

- **Pooling**: each slide has a **timestamp range.** Transcription sentences, slide bulletpoints get all grouped based off time stamp range = **segment**
    - Note blocks also grouped based off the timestamp range, but independently



as seen in the image, the slide's timestamps determine the information that will be compared. this works both for when we have slides + transcript or slides + notes. if the slides are not available, the transcription and the notes will be pooled in 4 minute comparison windows.

For each slide window, group overlapping transcript sentences and the slide's bullets. Retrieve note blocks independently, including nearby notes to allow for writing delay. - Initial example: for a 10:00-12:00 slide window, retrieve notes and transcript context from 10:00-13:00. Judge source items in the main window; use neighbors as context only.

MERGE PROCESS

- Jev (TypeSafe "System 1" thinking model) compares the **pool** of information (slides' bullets and/or transcriptions). against the notes of the same segment (a note block). it takes one pool point and asks if it's anywhere in the notes
    - returns one of: covered / partially covered / missing / noise
        - noise filters any remaining non-relevant section. it is ignored - ie even though that sentence is not covered in the notes, it is ignored. we need very specific indications of what is labeled as noise.
        - missing and partially covered -> will be edited later. stored.

- Jev also compares the notes against the pool of information. this time it takes one note block and asks if it conflicts any information in the pool. the question contains the note ID. important that the question contains the note ID, under the assumption that doing so allows the later llm model to know WHICH question returned conflicts. returns one of: supported / unsupported / conflicts.

        - only flags conflicts; adds it to edit list

- edit list: everything that was flagged alongside the ID. check if with the current infrastructure, the later llm model will know: exactly what is missing, exactly what conflicted.

- haiku model gets the all the to edit list:
    - missing items: locate best place and write addition
    - partial items: locate and add only what is missing
    - conflicting: locate, propose correction
    - all parts should be able to be cited -> ie the ID is kept.

- Repeated for every segment

HUMAN IN THE LOOP

- interface shows completed notes with additions and conflict flags:
    - additions are highlighted green blocks. default as “keep”, but if hover, widget appears above the text and shows the citation of source + the ability to “revert”
    - contradiction flags are yellow blocks. no default. hover, widget appears above text and shows the sources and proposed rewrite + the ability to “keep as is” or “edit” with proposed rewrite.
- selecting text allows me to hear it back with the corresponding audio (if available)